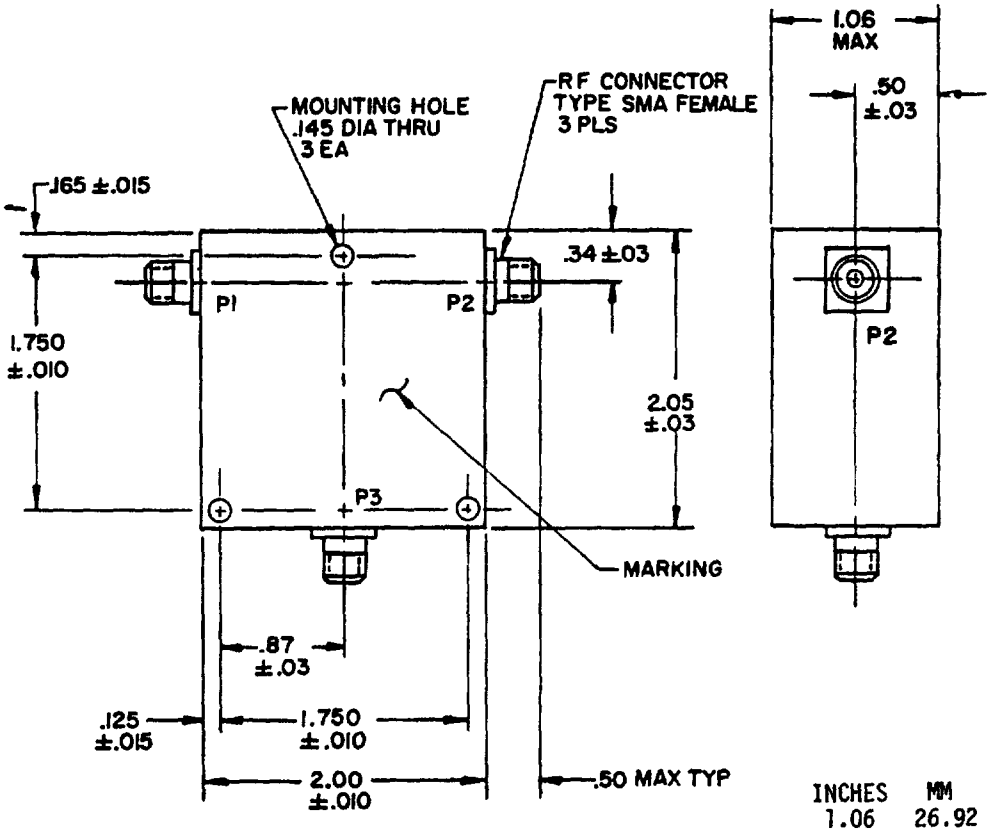
MILITARY SPECIFICATION SHEET

CIRCULATOR, RADIO FREQUENCY, COAXIAL,
(3 PORTS, FREQUENCY RANGE - .15 TO .18 GHz)

This specification sheet is approved for use by Naval Electronic Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the circulator described herein shall consist of this document and the latest issue of Specification MIL-C-28790.



INCHES	MM
1.06	26.92
2.00	50.80
2.05	52.07

NOTES:
1. Dimensions are in inches.
2. Metric equivalents (to the nearest .01 mm) are given for general information only and are based upon 1 inch = 25.4 mm.

FIGURE 1. RF coaxial circulator.

FSC 5985

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ENGINEERING DATA:

Frequency range: .15 to .18 GHz.
Nominal impedance: 50 ohms.
Operating temperature range: -50°C to +100°C.
Power rating:
Average - 10 watts.
Peak - 1 kilowatt.
VSWR maximum load: 2.5:1.
Circulation: 1 ⟶ 2 ⟶ 3 ⟶ 1.
RF connectors: Shall be stainless steel and shall mate with SMA male in accordance with MIL-C-39012.
Maximum altitude: 10,000 feet.

REQUIREMENTS:

Dimensions and configuration: See figure 1.
Dielectric withstanding voltage: 1,000 volts rms minimum.
Isolation:
-50°C to 0°C - 10 dB minimum.
+5°C to +70°C - 17 dB minimum.
+75°C to +100°C - 15 dB minimum.
Insertion loss:
-50°C to 0°C - 2 dB maximum.
+5°C to +70°C - 1 dB maximum.
+75°C to +100°C - 1.5 dB maximum.
VSWR:
-50°C to 0°C - 2.2:1 maximum.
+5°C to +70°C - 1.4:1 maximum.
+75°C to +100°C - 1.5:1 maximum.
Shock (specified pulse): In accordance with MIL-C-28790, except test condition H.
Part number: M28790/2-01.
NOTE: The circulator covered by this specification sheet is substitutable for the following commercial types:

Part number	Manufacturer's code	Commercial type
M28790/2-01	37695	368674-1
M28790/2-01	16453	32548

Custodian:
Navy - EC

Review activities:
Navy -
DSA - ES

User activity:
Navy - OS, AS

Preparing activity.
Navy - EC

Agent'
DSA - ES

(Project 5985-N453)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
No 22-R255

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DOCUMENT IDENTIFIER AND TITLE

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☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1 HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A GIVE PARAGRAPH NUMBER AND WORDING

B RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2 COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3 IS THE DOCUMENT RESTRICTIVE?

☐ YES ☐ NO *(If "Yes", in what way?)*

4 REMARKS

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